Preprocessing Steps & Business Questions with Visualization

Date	07 July 2026
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Phase	Project Development Phase

1. Preprocessing Steps

- Connected Tableau to Consumption.csv — 16,599 rows covering 33 states across 5 regions (NR, SR, ER, WR, NER), January 2019 to December 2020.
- Verified field types: Dates as Date, Usage as a decimal number, States/Regions as strings; confirmed Latitude/Longitude generate correctly for mapping.
- Checked for missing values — none found across any field in the source file.
- Standardized state name spellings so states group and filter consistently across worksheets.
- Identified a reporting-frequency gap: 2020 has far fewer recorded days per state on average than 2019 (≈144 vs ≈359 days/state). Comparisons that could be skewed by this gap use average daily usage rather than raw yearly totals.
- Created calculated fields: a lockdown-period flag, and Top N / Bottom N rank fields (RANK(SUM([Usage]))) driving the ranking views.

2. Business Questions with Visualization

Q1. How did India's overall electricity consumption trend change from 2019 to 2020?

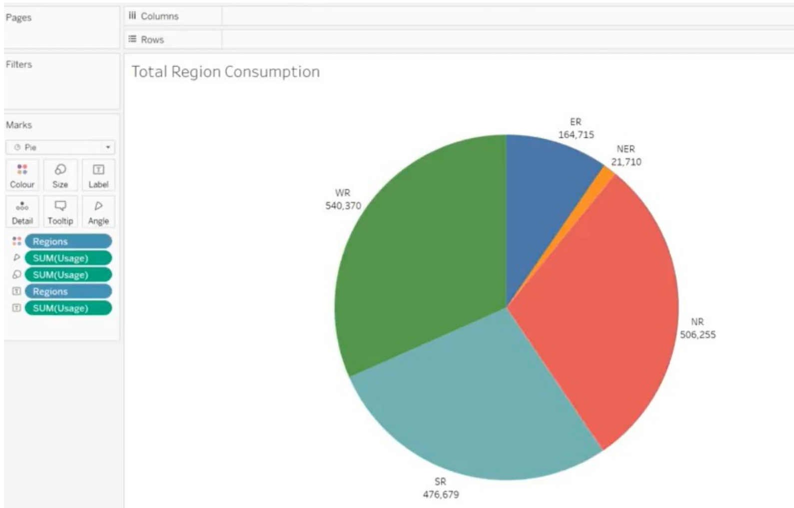
Visualization: Usage by Region — bar chart of total usage by region, split by year.

Usage by Region, 2019 vs 2020

Insight: Every region shows a lower 2020 bar than 2019 in raw totals (e.g., NR: 358,695 → 147,560 MU). This reflects both the lockdown period and the reduced number of reported days in 2020 noted above, rather than a proportional drop in daily demand alone.

Q2. How is total consumption distributed across regions?

Visualization: Total Region Consumption — pie chart of usage share by region.

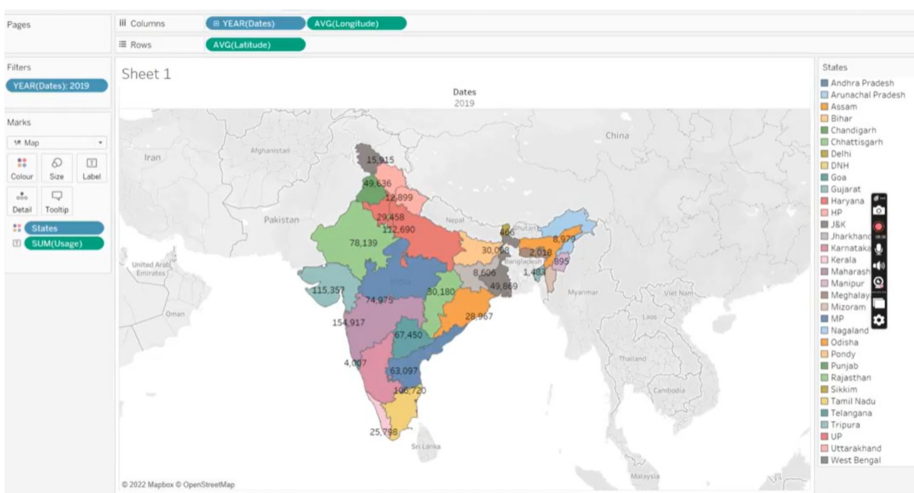


Total Region Consumption share

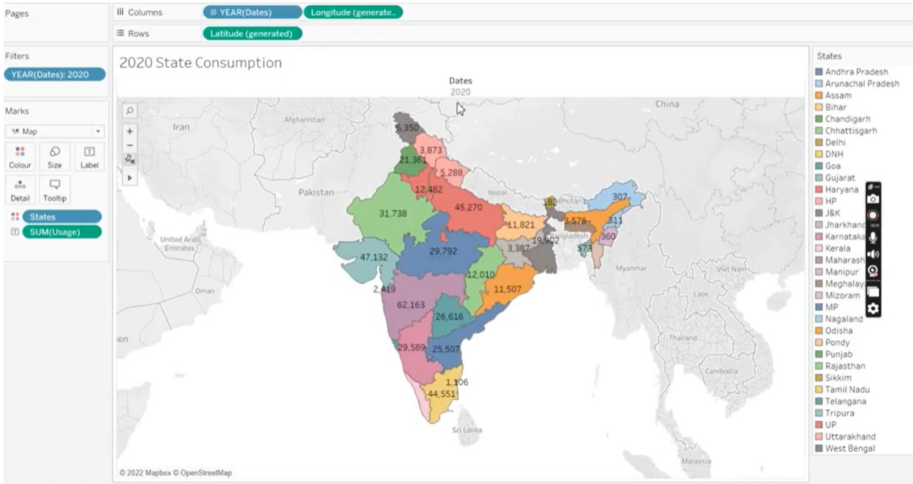
Insight: WR (540,370 MU) and NR (506,255 MU) together account for the majority of total recorded consumption, while NER (21,710 MU) is a small fraction — consistent with population and industrial concentration across regions.

Q3. How does consumption vary geographically by state, and how did it shift between 2019 and 2020?

Visualization: State Consumption maps for 2019 and 2020.



2019 State Consumption

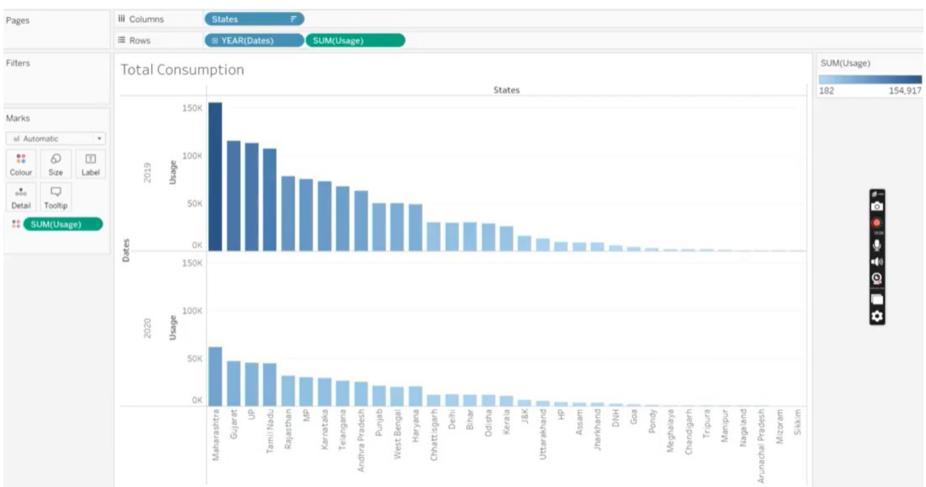


2020 State Consumption

Insight: Maharashtra, Uttar Pradesh, Gujarat, and Tamil Nadu remain the largest consumers in both years — the overall geographic concentration of demand holds steady even as absolute totals fall in 2020.

Q4. Which states are the top and bottom consumers, and how did rankings shift?

Visualization: Total Consumption — dual bar chart comparing each state's usage in 2019 vs 2020.



Total Consumption by state, 2019 vs 2020

Insight: Maharashtra leads in both years, followed by Gujarat, UP, and Tamil Nadu. The smallest states — Sikkim, Mizoram, Nagaland — remain the lowest consumers in both years, showing the state ranking is largely stable year over year.

Q5. Which states rank highest/lowest using a dynamic Top N / Bottom N view?

Visualization: Region Wise State Usage dashboard, combining a region/state usage table with Top N and Bottom N ranking charts.

Region Wise State Usage — Top N / Bottom N (N = 5)

Insight: With the Top N parameter set to 5, Maharashtra, Gujarat, UP, Tamil Nadu, and Rajasthan appear as the top consumers across both years, while the Bottom N view highlights the smallest North-Eastern states — giving planners a quick, adjustable way to focus on either end of the distribution.